Federated Learning Simulator for Medical Data Anonymization

Field:
AI & Machine Learning (Federated Learning, Privacy-Preserving AI)

Core Concept:
A simulator that demonstrates and evaluates Federated Learning (FL) for medical imaging tasks (e.g., classifying chest X-rays for pneumonia). The focus is on balancing three critical factors:

- **Model accuracy** (diagnostic performance)
- **Communication cost** (bandwidth efficiency)
- **Privacy guarantees** (via differential privacy)

Why It Stands Out:
Unlike traditional centralized AI, federated learning keeps sensitive medical data decentralized (at hospitals/clinics) while still enabling collaborative model training. This simulator lets you explore complex system dynamics without needing real-world hospital partnerships.

System Architecture & Technical Depth

1. Distributed Simulation Framework

- **Central Coordinator:** Manages model aggregation, client orchestration, and differential privacy enforcement
- **Client Simulators:** Multiple virtual "hospitals" running on a single machine, each with localized data
- **Communication Protocol:** Secure model update exchange (FedAvg algorithm)

2. Backend Implementation

- Client registration and authentication system
- Federated Averaging (FedAvg) implementation from scratch
- Differential Privacy integration:
    - Calibrated Gaussian noise injection into model updates
    - Privacy budget (ε) tracking and enforcement
- Model aggregation with robustness checks

3. AI/ML Core Components

- **Base Model:** Modified ResNet-18 for chest X-ray classification
- **Dataset:** Public medical imaging dataset (e.g., CheXpert or MIMIC-CXR)
- **Federated Training Loop:**
    - Local training at client nodes
    - Secure model update transmission
    - DP-noise addition before aggregation
    - Global model evaluation

4. Algorithmic Focus Areas

- Optimizing the FedAvg algorithm for medical data characteristics
- Tuning differential privacy parameters (ε, δ) for medical use cases
- Handling non-IID (Non-Independent and Identically Distributed) data distributions
- Communication-efficient training strategies

Research Potential & Academic Value

Key Investigations:

- **Accuracy vs. Privacy Tradeoffs:**
  How different DP noise levels (ε values) affect diagnostic accuracy
  Finding the "sweet spot" between privacy and performance
- **Communication Efficiency:**
  Comparing model convergence rates under different:
  - Update frequencies
  - Client sampling strategies
  - Compression techniques
- **Non-IID Data Challenges:**
  Simulating real-world medical data distributions where:
  - Some hospitals see mostly pneumonia cases
  - Others see mostly normal cases
  - Some have better imaging equipment than others

Evaluation Metrics:

- **Primary:**
  Model Accuracy (AUC-ROC for pneumonia detection)
  Privacy Budget (ε value from differential privacy)
  Communication Cost (MB transferred per training round)
- **Secondary:**
  Convergence Speed (rounds to reach target accuracy)
  Fairness (accuracy parity across client data distributions)
  Robustness (performance under client dropouts)

Visualization Opportunities:

- Pareto frontiers showing accuracy vs. privacy tradeoffs
- Convergence curves under different DP settings
- Communication cost breakdowns

Practical Implementation

Technology Stack:

- **Frameworks:** PyTorch or TensorFlow

- **Libraries:**
  PySyft or *TensorFlow Federated* for FL components
  TensorFlow Privacy for DP implementation
  Opacus (PyTorch DP library) as alternative
- **Data:** CheXpert or MIMIC-CXR datasets (publicly available)
- **Compute:** Runs on standard laptops or free Colab instances

Development Phases:

1. **Core FL System** (2-3 weeks)
   - Basic FedAvg implementation
   - Client-coordinator communication
   - Local training loops
2. **DP Integration** (2 weeks)
   - Noise injection mechanisms
   - Privacy budget accounting
   - Accuracy impact analysis
3. **Evaluation Framework** (2 weeks)
   - Metric calculation system
   - Visualization dashboards
   - Experiment automation
4. **Advanced Experiments** (1 week)
   - Non-IID data simulations
   - Communication optimization
   - Robustness testing

Why This Is Feasible:

- No special hardware required
- All components have existing open-source implementations
- Public medical datasets eliminate data collection hurdles
- Modular design allows incremental development

Potential Impact:
This simulator could serve as:

- A research tool for exploring FL-DP tradeoffs in healthcare
- An educational platform for medical AI privacy concepts
- A testbed for developing real-world federated medical systems

The work directly addresses critical challenges in medical AI:

- Patient privacy preservation
- Multi-institutional collaboration
- Resource-constrained training scenarios